Channel assignment and voice QoS in synthetic 802.11 networks: results

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All numbers below are produced by `paper/results.jl` on synthetic worlds (16 threads); the model and methods are described in the companion document `model.pdf`. A world places up to 18 access points (APs) at random, each with its clients, and an assignment picks one channel per AP. The throughput objective is proportional fairness, $\sum_c \log r_c$ over client rates r_c, measured in nats.

1 Channel assignment against the exact optimum

Each method is scored against the exact optimum from the subset dynamic program, on 270 worlds: 10, 14 and 18 APs, 4 and 16 clients per AP, ten seeds per setting for 2, 3, 4 and 6 channels at 40 Mbps demand, and five seeds for 3 channels at 3 Mbps ("light", where APs are not saturated and the airtime fixed point is active). These worlds use an area of $100\sqrt{n/18}$ m per side, 8 dB shadowing and a 0.001 Mbps rate floor; the voice scenarios use the default model. Annealing runs 50 000 moves.

Table 1: Share of worlds solved exactly (gap below 10^{-9} nats), by number of channels.

Method	2	3	4	6	3, light
Graph colouring	2 %	2 %	0 %	0 %	10 %
Greedy, single pass (original)	2 %	0 %	3 %	3 %	37 %
Greedy with local search	63 %	38 %	30 %	43 %	73 %
Annealing, fixed $T_0 = 5$, no polish	100 %	95 %	90 %	72 %	100 %
Annealing, calibrated T_0, polish	100 %	100 %	92 %	80 %	97 %
`optimize_assignment`	100 %	100 %	100 %	100 %	100 %

Table 2: Gap to the optimum in nats: median (worst), by number of channels.

Method	2	3	4	6	3, light
Graph colouring	27.56 (126.77)	40.94 (270.56)	67.49 (353.05)	95.56 (437.04)	38.17 (177.81)
Greedy, single pass (original)	10.30 (89.59)	12.88 (72.33)	11.48 (60.76)	6.09 (53.02)	2.98 (40.81)
Greedy with local search	0.00 (43.26)	1.12 (21.45)	1.20 (38.15)	0.34 (9.59)	0.00 (11.88)
Annealing, fixed $T_0 = 5$, no polish	0.00 (0.00)	0.00 (3.49)	0.00 (1.55)	0.00 (5.68)	0.00 (0.00)
Annealing, calibrated T_0, polish	0.00 (0.00)	0.00 (0.00)	0.00 (3.82)	0.00 (3.09)	0.00 (0.13)
`optimize_assignment`	0.00 (0.00)	0.00 (0.00)	0.00 (0.00)	0.00 (0.00)	0.00 (0.00)

`optimize_assignment` reaches the optimum in every world (worst gap 1.5×10^{-12} nats). The original single-pass greedy is 9.98 nats short in the median and up to 89.59 nats.

2 Speed

Table 3: Median wall time in ms, 16 clients per AP, 3 channels. "Original table" is the original single-threaded subset-table algorithm (same output to within 4×10^{-13}); it does not apply to light demand, which needs the per-group airtime fixed point. "Voice-aware" is the optimizer with the voice objective (64 calls per AP, 75 % background load).

Demand	APs	Original table	New table	Optimize	Exact	Voice-aware
heavy	12	18.3	6.3	7.6	6.3	11.1
heavy	14	93.3	6.1	14.0	11.4	35.1
heavy	16	401.2	29.4	24.8	24.2	130.7
heavy	18	1799.7	73.8	72.4	97.7	564.4
light	12	–	11.2	13.0	11.4	25.4
light	14	–	48.4	52.1	49.5	108.9
light	16	–	193.0	192.5	196.0	412.6
light	18	–	1769.7	1165.7	1449.1	2127.4

3 Model validation

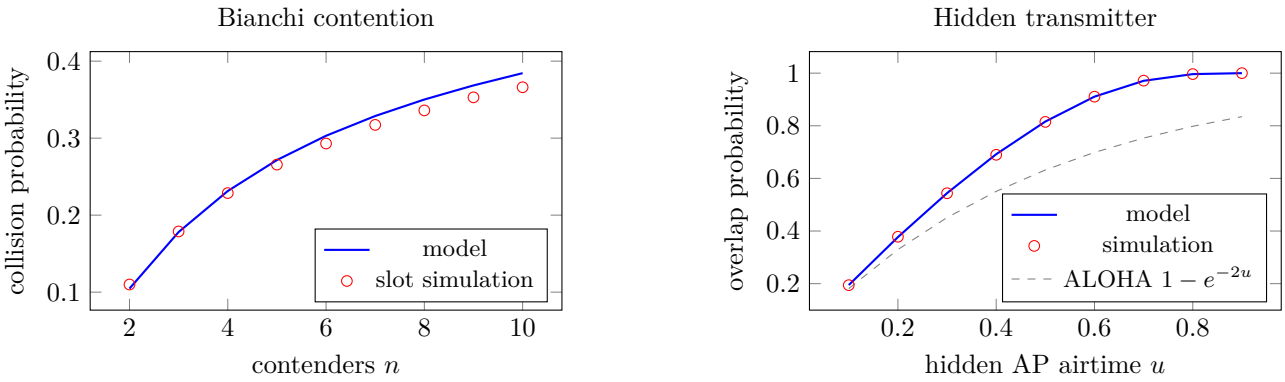


Figure 1: Collision models against simulation. The voice scenarios reach at most 4 contenders (the AP plus the co-channel APs it senses); up to that, Bianchi's fixed point is within 0.005 of a slot-level DCF simulation, and within 0.018 up to 10. The single-transmitter overlap formula is within 0.002; the ALOHA formula it replaced was off by up to 0.218.

Table 4: Steady-state voice estimate (used by the voice-aware objective) against `simulate_voice`: worst absolute error in loss (percentage points) / codec rate (kbps) over 20 assignments per cell.

Scenario	`:fifo`	`:deadline`	`:adaptive`	`:combined`
Branch office	0.00 / 0.00	0.00 / 0.00	0.00 / 0.00	0.00 / 0.00
Open-plan office	0.22 / 0.00	0.00 / 0.00	0.09 / 0.02	0.00 / 0.00
High-density call centre	0.49 / 0.00	0.15 / 0.00	0.32 / 0.07	0.00 / 0.01
Emergency overload stress	0.39 / 0.00	0.34 / 0.00	0.45 / 0.10	0.06 / 0.06

The queue simulator matches the closed-form steady state of an isolated AP in all 48 cases (loss within 0.17 points, codec within 0.03 kbps), and the voice-aware optimizer matches an exhaustive search of its objective in all 80 cases (worst gap 5×10^{-13}).

4 Voice

Table 5: Voice outcomes on throughput-optimal assignments, 20 worlds per scenario, 10 s simulated.

Scenario	Policy	Loss (median)	Loss (worst)	p95 delay (ms)	Codec (kbps)
Branch office	`:fifo`	0.01 %	0.65 %	7.8	64.0
	`:deadline`	0.01 %	0.65 %	0.4	64.0
	`:adaptive`	0.01 %	0.65 %	7.8	64.0
	`:combined`	0.01 %	0.65 %	0.4	64.0
Open-plan office	`:fifo`	0.16 %	2.10 %	17.5	64.0
	`:deadline`	0.16 %	1.03 %	1.8	64.0
	`:adaptive`	0.16 %	1.44 %	17.5	64.0
	`:combined`	0.16 %	1.03 %	1.8	64.0
High-density call centre	`:fifo`	11.73 %	19.31 %	140.0	64.0
	`:deadline`	1.19 %	3.45 %	8.5	64.0
	`:adaptive`	5.05 %	9.83 %	140.0	36.5
	`:combined`	1.19 %	3.45 %	8.5	64.0
Emergency overload stress	`:fifo`	40.85 %	46.89 %	140.0	64.0
	`:deadline`	5.55 %	15.09 %	140.0	64.0
	`:adaptive`	20.85 %	27.84 %	140.0	12.2
	`:combined`	0.53 %	2.90 %	37.6	59.6

Table 6: Throughput-only against voice-aware assignment (same policy for both): median loss, worlds where voice goodput rises / falls, and the cost to data clients in nats per client.

Scenario	Policy	Loss before	Loss after	Goodput up/down	Cost (median)	Cost (max)
Branch office	`:fifo`	0.01 %	0.01 %	1 / 0	0.000	0.000
	`:deadline`	0.01 %	0.01 %	1 / 0	0.000	0.000
	`:adaptive`	0.01 %	0.01 %	1 / 0	0.000	0.000
	`:combined`	0.01 %	0.01 %	1 / 0	0.000	0.000
Open-plan office	`:fifo`	0.16 %	0.08 %	2 / 0	0.000	0.034
	`:deadline`	0.16 %	0.16 %	0 / 0	0.000	0.000
	`:adaptive`	0.16 %	0.06 %	3 / 0	0.000	0.034
	`:combined`	0.16 %	0.16 %	0 / 0	0.000	0.000
High-density call centre	`:fifo`	11.73 %	8.42 %	17 / 0	0.039	0.092
	`:deadline`	1.19 %	0.58 %	9 / 0	0.000	0.033
	`:adaptive`	5.05 %	2.17 %	19 / 0	0.084	0.181
	`:combined`	1.19 %	0.58 %	9 / 0	0.000	0.033
Emergency overload stress	`:fifo`	40.85 %	40.36 %	16 / 0	0.010	0.042
	`:deadline`	5.55 %	3.38 %	19 / 0	0.034	0.078
	`:adaptive`	20.85 %	19.79 %	16 / 0	0.023	0.076
	`:combined`	0.53 %	0.65 %	19 / 0	0.032	0.088

Paired sign tests over 50 worlds: deadline scheduling beats adaptation alone on loss in 49 worlds (1 tied, 0 lost; median 3.12 points, $p = 1.8 \times 10^{-15}$) and cuts p95 delay in 50 (median 136.40 ms, $p = 8.9 \times 10^{-16}$). Under voice overload, priority with codec adaptation beats fixed-rate priority in 50 of 50 worlds (median 27.19 points, $p = 8.9 \times 10^{-16}$) at a median codec reduction of 26.1 kbps, and keeps at least 99 % of the codec rate in 200 of 200 normal-load trials.

5 Limits

Worlds, traffic and shadowing are synthetic and static. The voice model is an aggregate work queue, not a packet-level 802.11 simulator: Bianchi treats partly busy APs as fractional contenders, and each AP's own clients count as one contender, which likely understates collisions in call-heavy cells. The exact methods cover up to 18 APs. External validation needs ns-3, packet traces or hardware.